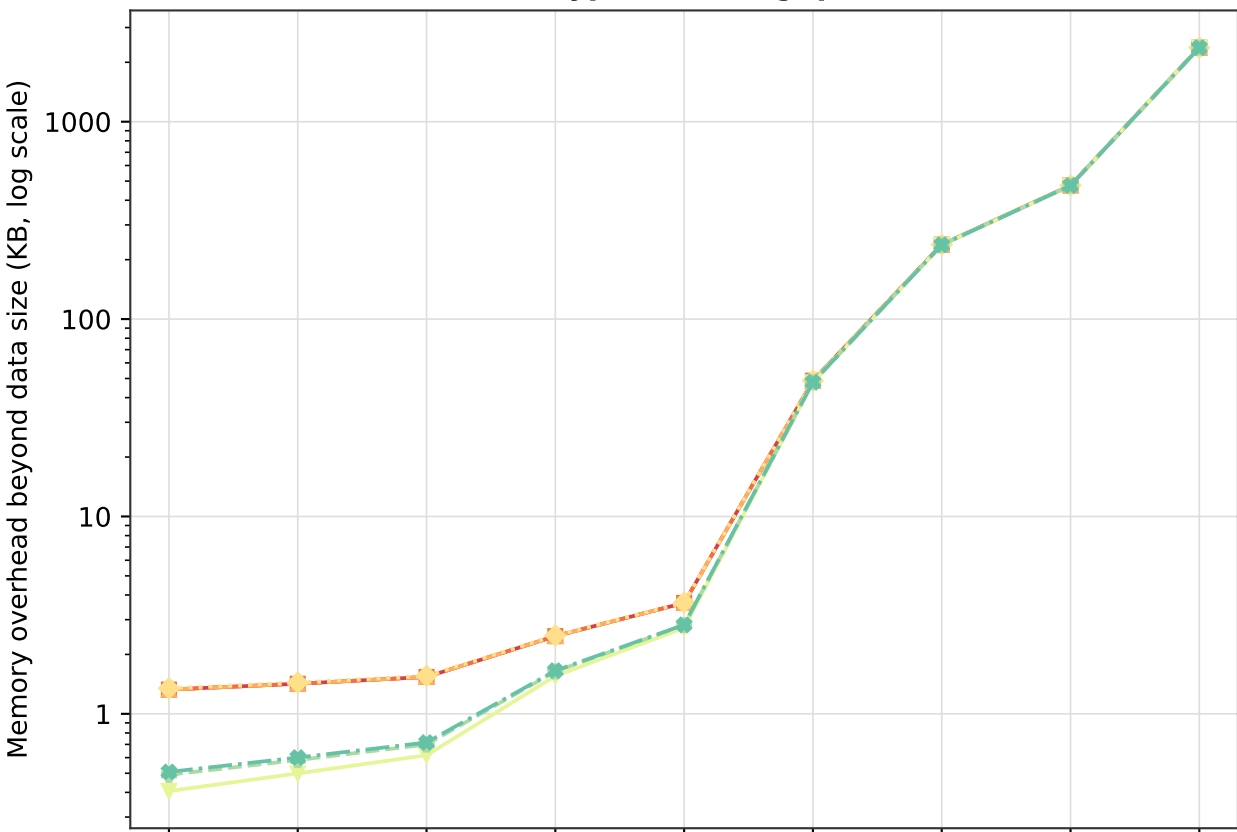


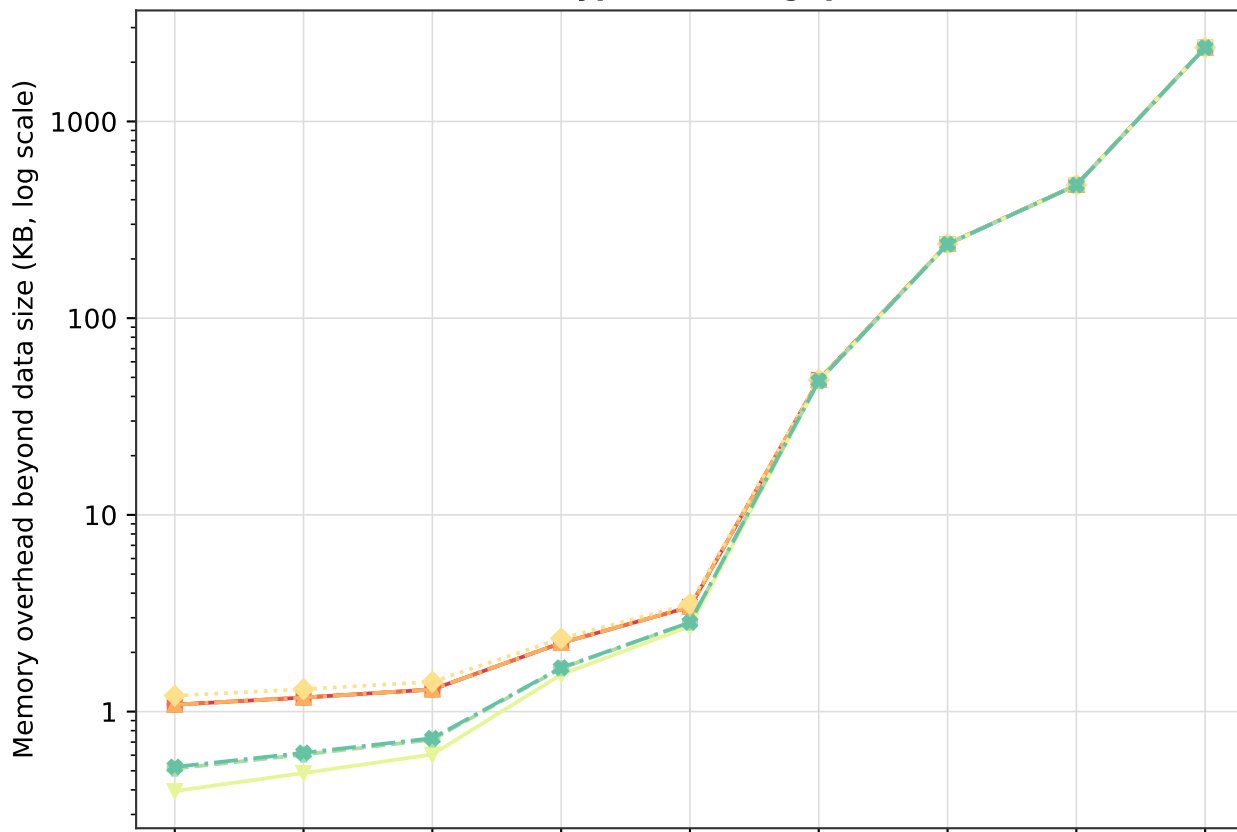
Phase 1 — Encryption vs. decryption throughput by cipher and packet size

AES-128 AES-192 AES-256 ChaCha20 HIGHT RECTANGLE SPECK

Encryption throughput



Decryption throughput



	1KB	5KB	10KB	50KB	100KB	1MB	5MB	10MB	50MB
AES-128	1.33	1.42	1.54	2.48	3.65	48.74	238	476	2,373
AES-192	1.33	1.42	1.54	2.48	3.65	48.74	238	476	2,373
AES-256	1.33	1.42	1.54	2.48	3.65	48.74	238	476	2,373
ChaCha20	1.35	1.44	1.56	2.50	3.67	48.76	239	476	2,373
HIGHT	0.406	0.499	0.617	1.55	2.72	47.82	238	475	2,372
RECTANGLE	0.491	0.583	0.699	1.64	2.81	47.90	238	475	2,372
SPECK	0.508	0.601	0.716	1.65	2.83	47.92	238	475	2,372

	1KB	5KB	10KB	50KB	100KB	1MB	5MB	10MB	50MB
AES-128	1.08	1.18	1.30	2.23	3.41	48.50	238	475	2,373
AES-192	1.08	1.18	1.30	2.23	3.41	48.50	238	475	2,373
AES-256	1.08	1.18	1.30	2.23	3.41	48.50	238	475	2,373
ChaCha20	1.20	1.30	1.42	2.35	3.53	48.62	238	476	2,373
HIGHT	0.395	0.487	0.605	1.54	2.71	47.81	238	475	2,372
RECTANGLE	0.513	0.604	0.721	1.66	2.83	47.93	238	475	2,372
SPECK	0.523	0.617	0.732	1.67	2.84	47.94	238	475	2,372